

KNOTBOOK-An Online Learning Platform

Gokul G Nair
Computer Science and Engineering Dept.
Mar Baselios of Technology and Science (MBITS)
Kothamangalam, Kerala, India
gokulgnair2003@gmail.com

Solaman Aji
Computer Science and Engineering Dept.
Mar Baselios of Technology and Science (MBITS)
Kothamangalam, Kerala, India
solamanaji@gmail.com

Nisreen M Ali
Computer Science and Engineering Dept.
Mar Baselios of Technology and Science (MBITS)
Kothamangalam, Kerala, India
line 5: email address or ORCID

Sheena S Saiju
Computer Science and Engineering Dept.
Mar Baselios of Technology and Science (MBITS)
Kothamangalam, Kerala, India
sheenasaiju5@gmail.com

Susan Paul
Computer Science and Engineering Dept.
Mar Baselios of Technology and Science (MBITS)
Kothamangalam, Kerala, India
susanpaulmambillil@gmail.com

Abstract— Knotbook is an online learning platform designed to streamline the process of study material-sharing by integrating several AI features such as automatic question generation from uploaded documents, automatic evaluation and progress reporting. The platform allows teachers to upload study materials in PDF format. Using advanced natural language processing (NLP) techniques, the system automatically extracts key concepts, terminologies and other important information from the documents. Based on this extracted content, the platform generates a variety of practice questions, including multiple-choice, true/false and subjective format questions. This automation helps in creating a rich repository of practice material and assessments for students without requiring manual input. The platform aims to enhance the efficiency of

educational resource management and improve student engagement by providing a transformative, seamless and interactive learning experience for both students and the teachers. Progress reporting in the system helps to monitor student performance, engagement, identify at-risk learners and suggest intervention strategies to improve their learning. This can help educators intervene early, customize learning experiences and improve overall educational outcomes.

Keywords— *Material Sharing, Automatic Question Generation, Automatic Evaluation, Progress Reporting, At-Risk Learners*

I. INTRODUCTION

Digital learning has transformed education but struggles with the automation of key tasks like creation of exams and monitoring progress of

students. Building on a study material sharing platform, this project aims to introduce an advanced system for integrating content distribution, automated assessments and progress reporting.

The platform simplifies the processes like question generation, grading and performance tracking. The system helps in enhancing teaching efficiency and student engagement rate. The platform has a user-friendly interface that unifies essential tools for efficient workflow. The system is flexible and suitable to various needs, by supporting both blended and traditional learning and teaching methods while promoting innovation in education.

II. BACKGROUND

A. MCQGen: A Large Language Model-Driven MCQ Generator for Personalized Learning

MCQGen, also referred to as A Large Language Model-Driven MCQ Generator for Personalized Learning, is a sophisticated framework that automatically creates excellent multiple-choice questions (MCQs) that accommodate a variety of learning styles. This system uses Retrieval-Augmented Generation (RAG) in conjunction with GPT-4, a state-of-the-art large language model (LLM), to include external knowledge and create context-rich, educationally relevant queries. In order to improve logical depth and self-refinement—an iterative process that consistently improves question quality—it integrates advanced prompt engineering

techniques including Chain-of-Thought (CoT) reasoning. Crowdsourced evaluations generate a feedback loop that allows the system to adapt dynamically to improve its outputs and better meet the unique needs of learners. This system's revolutionary methodology not only creates multiple-choice questions (MCQs) to evaluate knowledge comprehension. The study benefits from an effective feedback mechanism and structured guidance for the LLM, ensuring continuous improvement and alignment with learning objectives. However, it faces challenges such as limitations in math-based questions, restricted question diversity and complexity, dependence on instructor input, and a limited subject domain.

B. UniRaG: Unification, Retrieval, and Generation for Multimodal Question Answering With Pre-Trained Language Models

The UniRag framework, "Unification, Retrieval, and Generation for Multimodal Question Answering with Pre-Trained Language Models," is an innovative approach to multimodal question answering (MMQA). It unifies text, image, and table data into a unified representation, minimizing information loss and enhancing transformation. This process involves linearizing all modalities via LLaVA and then retrieving context using a cross-encoder to identify relevant documents. The best k documents are chosen for answer generation, which employs a Flan model to generate accurate answers. The framework further enhances model differentiation by including distractor documents, thereby making it

more robust and adaptive. Although UniRag has been successful, it suffers from high computational complexity, information overload, and conversion errors. Its experiments have achieved state-of-the-art performance in MMQA, providing rich descriptions of images and maintaining tabular information without time limits or constraints.

C. Opinierium: Subjective Question Generation Using Large Language Models

This study focuses on developing subjective questions from news media posts to boost audience interaction. Unlike prior research that emphasized objective, fact-based questions, this approach generates subjective questions to elicit opinions. The methodology involves fine-tuning multiple *flan-T5* models and GPT-3 architectures using Seq2 sequences for conversational generation as a unified task. Testing was conducted on 40,000 news articles using self-generated and human-sourced questions to evaluate the model's effectiveness. The research compares fine-tuned models with GPT-3.5 under zero-shot prompting, highlighting differences in handling subjective question generation. The study addresses challenges in generating opinion-based questions, which are inherently open-ended and complex. A detailed analysis compares transformer architectures, using metrics like BLEU, ROUGE, METEOR, and BERTScore, to assess model performance. This approach advances the automatic generation of opinion-driven questions, contributing valuable

insights into subjective question generation and evaluation.

D. Leveraging Structured Information from a Passage to Generate Questions

This system introduces methodologies to improve automatic question generation (QG) by integrating structured information such as named entities, relationships, and events. Traditional QG methods often rely on unstructured text, resulting in generic or shallow questions. By combining unstructured passage text with structured extracted data through an entity and event extraction-based neural network model, the proposed approach generates diverse and contextually relevant questions. Key entities and their relationships are analyzed as inputs to the model, enabling the generation of questions that better reflect the content of the passage. Experimental results demonstrate that this structured model outperforms traditional QG methods in relevance, diversity, and alignment with the passage's content. The questions produced are more varied, informative, and context-aware. This approach is ideal for educational tools, automated content generation, and conversational AI, providing higher-quality output. By integrating structured data, the method significantly enhances the quality and utility of generated questions, making it a valuable contribution to context-aware QG systems.

E. A study of an Intelligent Algorithm for

Automatic Test Paper Generation and scoring in University English Exams

The system proposes an advanced automatic test paper generation that uses an Improved Genetic Algorithm (IGA) to increase the effectiveness and fairness of English tests. The model generates test papers within 25 seconds, which is a faster rate compared to traditional methods such as Genetic Algorithms and Particle Swarm Optimization. It addresses crucial test elements including category, type, score, and difficulty while ensuring the validity of the test paper. The system also has an automatic scoring system for subjective questions, which includes grammar error detection and cosine similarity for scoring, all over a correlation of over 0.8 between manual and automated scores. This proves that the application is reliable, objective, and consistent. The application of this system enhances security, fairness, and scalability for large-scale English exams, making it applicable for educational evaluations. By using Enhanced Genetic Algorithm (EGA), the system ensures a balanced mix of difficulty and score distribution, while automatic scoring reduces subjectivity in translation question evaluation. This approach holds the potential to revolutionize large-scale English exams, promoting efficiency, transparency, and faster management in educational institutions worldwide. The system's innovative nature ensures consistent, fair, and efficient assessment, benefiting both domestic and international educational systems.

F. Automatic Exam Correction Framework for the MCQs, Essays and Equations Matching

This system proposes an integrated approach to automate responses for various exam types, including multiple-choice questions (MCQs), essays, and mathematical equations. Advanced Natural Language Processing techniques are used for grading essays, while symbolic computation evaluates mathematical equations. For MCQs, a traditional machine learning model matches student responses with predetermined correct answers. Essay grading employs deep learning techniques like recurrent neural networks or transformers, assessing content quality, coherence, and relevance based on predefined rubrics. For equations, symbolic matching ensures syntax and semantic accuracy. The system preprocesses input responses, extracts relevant features, and applies evaluation models tailored to each question type. Experimental results demonstrate the system's ability to improve accuracy and efficiency compared to traditional grading methods. It ensures high accuracy in grading MCQs, provides meaningful essay assessments, and evaluates mathematical expressions effectively. This framework is a robust tool for modern educational settings, offering scalable and reliable automated grading for large-scale assessments.

G. Toward Robust Evaluation: A Comprehensive Taxonomy of Datasets and Metrics for Open Domain Question Answering in the Era of Large Language Models

This system explores the issues and evaluation methodologies of ODQA systems, specifically large language models (LLMs). It provides a taxonomy to denote the current datasets and evaluation metrics, thereby helping to standardize assessments regarding ODQA systems. Particular key technologies associated with this system are NLP, Information Retrieval (IR), and deep learning techniques for improving performance in ODQA systems. It is on large pre-trained models such as GPT and BERT, fine-tuned for cross-domain answer generation. ODQA is analyzed by benchmarks categorized into data type: text vs. knowledge graphs and task type: factual vs. reasoning. Evaluation metrics focus on accuracy, relevance, and coherence while paying attention to how the models deal with question ambiguity, factuality, and response complexity. The findings point out that diversity, representative datasets and sound metrics for nuanced aspects of ODQA performance are required. The metrics consider not just answer correctness but ambiguity and complexity. This analysis requires more extensive sets of benchmarking practices that should direct the development of stronger ODQA systems.

H. Automatic Question Generation from Given Paragraph

Advanced Natural Language Processing (NLP) methods have been used in the creation of AQG, it focuses on transformer models for question generation from single paragraphs. The methodology utilizes BERT (Bidirectional Encoder Representations from Transformers) and

GPT (Generative Pre-trained Transformer) models, which are further trained to do these tasks. So that the resultant questions are syntactically correct and contextually appropriate, these models are first tailored to the task. In this way, the process begins with converting the input text into a more suitable computer-readable format called preprocessing and continues with tokenization, recognition of named entities (NER) and parts of speech tagging (POS), which are aimed at understanding the core concepts. The focus then shifts to generating the questions. After having identified the focus points of the paragraph, transformers, and in particular sequence-to-sequence models, can be used for this task. This ensures that the essential information is retained in the generated sentences. Most importantly, the generated questions are analyzed afterwards using metrics such as BLEU (Bilingual Evaluation Understudy) and ROUGE (Recall-Oriented Understudy for Gisting Evaluation). In this way, AQG is able to fully exploit the advantages of transformer models to fully automate question generation for learning purposes and in the process, increase relevance and coherence.

I. Investigating Educational and Non educational Answer Selection for Educational Question Generation

The system for automatic educational question generation system uses machine learning models to classify answers as educational or non-educational. It utilizes supervised learning

algorithms, such as Support Vector Machines (SVM), Random Forests, and Neural Networks, on labeled datasets for classification. Feature extraction is carried out using semantic analysis, keyword extraction, and sentence embedding models like BERT to capture contextual relevance. The system also incorporates context-based models, utilizing Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks to improve answer selection by considering surrounding context and educational objectives. Attention mechanisms are used to rank important text components, enhancing the question generation process. This approach ensures that selected answers contribute to generating educationally meaningful questions. The methodology is based on improving the accuracy and quality of the generated questions, which makes the system more effective for educational content creation. These findings are valuable for developing personalized learning platforms and intelligent tools for automatic question generation in educational environments.

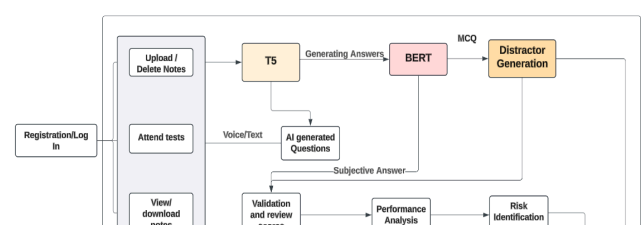
J. A Systematic Review of Automatic Question Generation for Educational Purposes

Automatic Question Generation (AQG) systems reduce the time and resource-demanding workload of creating appropriate questions on manual grounds. These include supporting assessments, personalized learning, and training AI models for tasks such as question-answering and machine reading comprehension (MRC) through extracting key information from text

using tokenization, part-of-speech tagging, and dependency parsing. Questions are generated using pre-defined templates and can be adapted to formats such as MCQs or subjective questions. Distractor generation techniques produce plausible incorrect options, and difficulty levels match the learner's capabilities. Although AQG systems provide efficiency, scalability, quality, and cost-effectiveness, they have limitations. They rely on rule-based templates, which limit linguistic variety and innovation. They also fail to generate complex, higher-order questions that require inference and critical thinking and do not have features for giving detailed feedback on errors. Despite these challenges, AQG systems enhance educational experiences by automating question generation and adapting in real-time to learners' needs, with potential for improvement in diversity and critical content generation.

III. SCOPE AND OBJECTIVE

The proposed system makes it easier for teachers and learners to interact with study materials, providing a platform that helps faculties to upload, organize, and share resources with efficiency. The goal of the system is to make learning more attainable and interesting by offering a simple and organized way for students to access study materials. This platform is centralized in nature, which increases the availability of content that students need, thereby facilitating a more connected and organized learning environment.



One of the key features of the system is the automatic generation of online exams based on the study materials uploaded by faculty. This ensures faster and more efficient development of assessment tests. The system intake content provided by the faculty and creates dynamic, relevant, customized exams that include a variety of question formats like MCQs and subjective questions. This ensures that assessments are relevant to what students have been learning and makes it easier for faculty to manage the exam process. With this automation, both faculty and students benefit from a time-efficient, accurate and smoother experience. The system also focuses on inclusivity, incorporating a Text-to-Voice conversion tool to ensure that all students, including those with visual impairments or other learning challenges, can fully participate in exams and access course materials. This feature provides equal opportunities for students who need additional support, making the learning experience more equitable. In addition, the system uses analytics to track student progress, identifying those who may be at risk of falling behind. By generating alerts for faculty and academic counselors, the system ensures that timely support is provided to students who need it most, helping them stay on track and succeed.

system provides a comprehensive solution to help both faculty and students thrive in today's educational landscape.

The proposed system architecture of Figure 1 below illustrates the suggested system design. It consists of an AI-driven framework designed to process user-uploaded content (e.g., PDF notes) and facilitate question generation, performance analysis and risk identification.

Fig. 1. Architecture Diagram

The T5 model scans the uploaded notes to formulate questions and accompanying answers that contextually fit the user's study materials, which will be placed in tests. On the other hand, the BERT model is used to provide more answers and valid distractors for the multiple choice questions. To enable the system to achieve its goals, a set of procedures are developed and incorporated. From the uploaded notes, the T5 model will create the AI generated question and the BERT model will work in parallel to create distractors that are competent and realistic. The answers are validated, the scores reviewed and the results are stored in a centralized secure database to maintain the confidentiality of the users. The overall functionality of T5 model is shared below in figure 2.

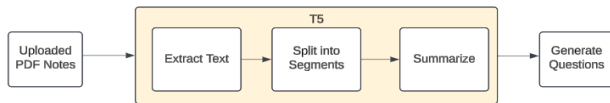


Fig. 2. T5 Model

The system allows users to upload their PDF notes, which are the primary input for the T5 model's workflow. After uploading, the system processes and extracts the text content from the PDFs, preparing it for further analysis. The extracted text is then divided into smaller, manageable segments to facilitate efficient processing and ensure that no critical information is overlooked. Once the data are segmented, this T5 model produces an abstraction of a paragraph in three short sentences so as to present an overview and streamline the generation process of the question. In short, relevant meaningful

questions are produced in coherence with these overviews in place of irrelevant answers. With all these workflows placed in proper succession, a product of great, contextually right questions ensures and enhances better output in the overall learning curve.

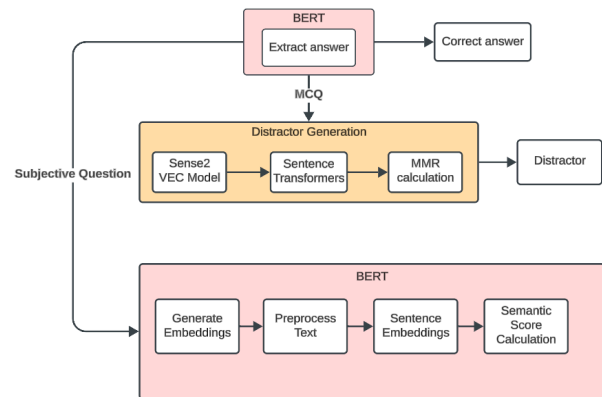


Fig. 3. BERT Model

The figure 3 details the BERT model which uses an advanced approach for enhancing the generation of multiple-choice questions (MCQs) by integrating advanced models of natural language processing. Firstly, BERT, which stands for Bidirectional Encoder Representations from Transformers, is used to extract the correct answers from subjective questions. It ensures that the answers are contextually accurate and relevant. To produce plausible distractors, the incorrect options that will be contextually similar to the correct answer, the system would use Sense2Vec, that identifies word level contextual relationships and Sentence Transformers to embed sentences into capturing semantic understanding. This way it is possible to create relevant as well as

challenging distractors.

To keep maintaining the quality of these distractors, the system computes MMR values. MMR ensures the distractors are relevant to the context but yet distinct enough from the correct answer, thereby enriching the quality of the MCQs. Furthermore, BERT computes the semantic similarity of possible answers with distractors for logical consistency and coherence in the set of questions. In this holistic methodology, the resulting MCQs are challenging as well as educationally worthwhile.

V. PROPOSED METHODOLOGY

An integrated, automated learning platform for the management of educational content, assessments, and student progress will be proposed. Faculty will upload and process study materials, automatically generate questions, provide customized examinations, and evaluate student performance efficiently, benefiting both educators and learners. It also comprises the Document Upload and Processing Module and the AQG Module, which facilitate the smooth translation of content into accessible formats, with assessments targeted toward specific course objectives. Another salient feature of this platform is its use of current technologies, including NLP and ML, to power its components. It is this module that will use the developed technologies to automatically generate a range of question types, such as multiple-choice, fill-in-the-blank, and subjective questions, from uploaded content. Faculty will review, edit or accept these questions,

ensuring that the generated assessments are academic in quality and relevance. Furthermore, the Examination Generation Module allows for the customization of exams so that instructors can create their own assessments based on the difficulty level, time limit, and even randomization, ensuring that each student is assessed fairly.

The Automatic Evaluation and Progress Report Generation Modules make the platform more efficient by immediately providing feedback about student performance. The evaluation module automatically grades objective questions and gives preliminary scores for subjective ones, thus saving faculty workload. Meanwhile, the feature of generating progress reports tracks and visualizes student performance over time, helping both students and faculty understand strengths and areas for improvement. Together, automation, customization and real-time feedback create a complete and engaging learning environment that allows for continuous student growth while relieving the administrative burdens on educators.

A. Automatic Question Answer Generation

This module automates the generation of questions and answers from a PDF document, supporting both MCQs with distractors and subjective questions. After a user uploads a PDF, the PyPDF2 library extracts raw text, which is divided into smaller segments for easier processing. The T5-small model summarizes the content into concise key points, which are then passed to the T5-base model to generate

multiple-choice and subjective questions based on the summaries. The BERT model extracts accurate answers for each question by analyzing the summarized text. Generated questions and answers are stored in a database for easy management and retrieval. To create distractors for MCQs, the Sense2Vec model identifies semantically similar words or phrases, while Sentence Transformers generate embeddings to ensure contextual relevance. The correct answer and distractors are combined to form balanced, contextually accurate multiple-choice options. This streamlined approach ensures high-quality question generation, enabling efficient and organized assessment creation.

B. Text-Voice Conversion

The Text-to-Speech (TTS) engine converts written content into natural-sounding spoken audio, enabling users to hear information rather than read it on-screen. This technology, integrated into platforms via TTS services such as Google Text-to-Speech, Amazon Polly or Microsoft Azure Speech, offers valuable functionality for users who are visually impaired, those who prefer auditory learning, or individuals multitasking with hands-free access. TTS can be customized with voice options, accents and language preferences, enhancing accessibility and allowing developers to create engaging, user-friendly experiences across educational, entertainment and productivity applications. The Speech-to-Text (STT) engine enables voice-based inputs to be transcribed into written text, allowing users to speak commands, dictate notes, or communicate more naturally with

applications. Integrated STT services such as Google Speech-to-Text, IBM Watson Speech to Text, and Apple's Dictation API use machine learning and NLP to accurately convert spoken language into text, supporting multiple languages and dialects. This feature is essential for applications aiming to increase accessibility, especially for users with physical limitations and for productivity tools that benefit from hands-free operation, such as virtual assistants, customer service bots and voice-activated search systems.

C. Answer Validation

In the Answer Validation Module, it primarily focuses on automatic grading of students' responses with a score, according to how correct their answers are. Automatic Grading validates and scores the students' answers based on their comparison with the correct answers saved in the database. The system checks for correctness in every response by making use of techniques such as semantic similarity and exact match, depending on the type of question. It gives scores to every student's response based on the degree of correctness. For example, a student may be given a high score if his or her answer is very similar to the correct answer or lies within an acceptable range of similarity. In cases where the answers were partially correct, the system could give the scores accordingly. Again, the application of MMR indirectly supports grading. Although MMR is primarily used in the question generation module to select suitable distractors for multiple-choice questions, it improves the quality

of the MCQ options, which makes it easier to evaluate whether students have selected the correct answers with genuine understanding rather than random selection.

D. Progress Reporting

This study explores an integrated approach to enhancing student performance through continuous tracking, in-depth analytics and intuitive progress monitoring. The system continuously logs student activities, such as assignment completion, quiz scores and discussion participation, allowing educators to identify patterns in performance and areas needing improvement. This methodology not only employs student accountability but also provides valuable observations for teachers, students and parents to adjust and plan their learning strategies. Through performance analytics, the system conducts deeper analyses of student data, revealing trends in grade progression, skill and development. This system will help in identifying students at risk or those ready for advanced content, enabling targeted interventions. The progress dashboard aggregates key performance indicators, visually presenting test scores, assignment statuses and participation rates in an easily understandable format, helping students, parents and educators quickly assess academic standing. This comprehensive system empowers students to set goals and reflect on progress, while providing educators with the tools to monitor performance efficiently and make timely adjustments. Collectively, these features support

personalized learning, facilitate responsive teaching and improve educational outcomes.

VI. RESULT AND DISCUSSION

The proposed system effectively generates high-quality questions and distractors from documents using BERT, Sense2Vec, and T5 models. It outperforms baseline and pre-trained models in accuracy, precision, and ROUGE scores. BERT ensures contextual relevance, Sense2Vec enhances distractor quality, and T5 generates clear, grammatically correct questions, validated through human evaluation.

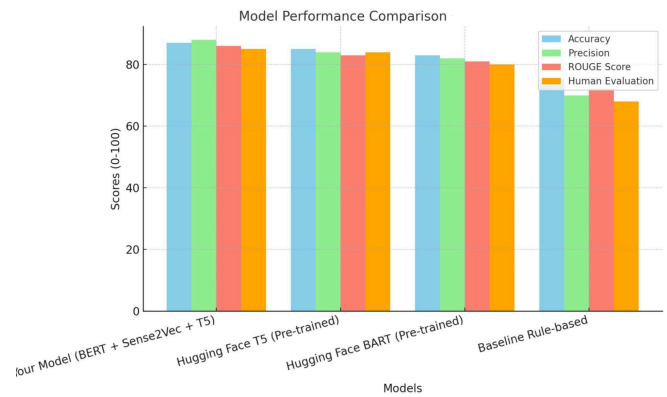


Fig. 4. Model Performance

The bar graph in figure 4 compares models for MCQ generation based on Accuracy, Precision, ROUGE Score, and Human Evaluation (0–100 scale). Models include the proposed T5 + BERT + Sense2Vec, Hugging Face T5, BART, and a Baseline Rule-based system. All models scored between 80–90, with the proposed model excelling in ROUGE Score and Human Evaluation, showcasing its strength in generating high-quality outputs. Hugging Face models performed consistently, slightly outperforming the

rule-based baseline. This demonstrates the proposed model’s effectiveness in domain-specific tasks.

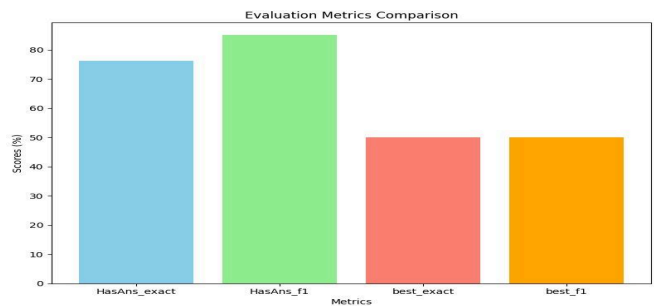


Fig.5.BERT model after fine tuning

The evaluation results in figure 5 showcase how well the fine-tuned BERT model performs in answer extraction. Key metrics include HasAns_exact at 76.21% and HasAns_f1 at 84.99%, demonstrating solid performance in extracting answers when they are present. The overall F1 score stands at 42.43%, with the highest F1 score reaching 50.07% at a threshold of 0. These findings emphasize both the model's strengths and the areas that need enhancement.

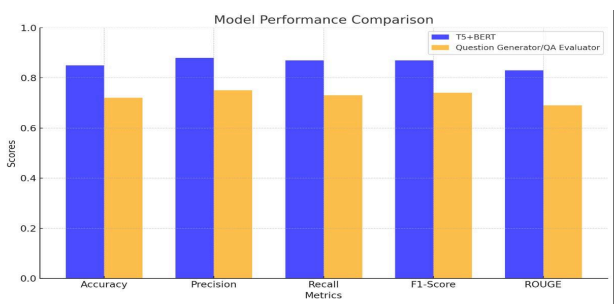


Fig 6. Model performance

The bar graph in figure 6 compares the performance of the T5+BERT model against the

Question Generator+QA Evaluator for subjective question-answer generation and analysis. Metrics assessed include Accuracy, Precision, Recall, F1-Score, and ROUGE. The T5+BERT model consistently outperforms the Question Generator+QA Evaluator across all metrics, demonstrating higher reliability and precision. Notable differences in F1-Score and ROUGE highlight T5+BERT’s strength in generating relevant and high-quality outputs, making it more effective for subjective evaluation tasks compared to the alternative approach.

VII. CONCLUSION

The online learning platform addresses the challenges of modern digital education by integrating AI-driven automation for text processing, exam generation, evaluation and progress reporting. The system enhances both teaching and learning experiences. With NLP-based machine learning models, the system automates question generation by ensuring diverse, relevant questions aligned with study materials, while customizable exam options like question type selection make the system more user-friendly. Automated grading of objective and subjective questions significantly reduce grading workloads, providing accurate and timely feedback. Faculty can oversee the process to ensure fairness, making the platform ideal for managing large amount of data for big classes. By merging automation with manual oversight, it put forwards a flexible, effective and student-centered

educational environment, empowering educators and supporting students with opportunities for personalized growth. This innovative approach sets a benchmark for accessible, efficient and modern digital education.

Future enhancements for the online learning platform could make it more interactive, adaptive and inclusive, enriching the educational experience for both students and educators. Attention-tracking technology could monitor student focus during lessons, allowing faculty to regulate pacing and delivery for better engagement, while sentiment analysis could encourage a supportive learning environment by interpreting student emotions. Adaptive learning modules could personalize education by adjusting content difficulty and recommending resources based on individual performance and gamification elements like rewards and leaderboards could boost motivation, making learning more interactive and goal-oriented. These advancements would support diverse learning styles, enhance outcomes and solidify the platform's position as a leader in educational technology.

REFERENCES

- [1] Ching Nam Han, Chee Wei Tan, Pei Duo-You, "MCQGen: A Large Language Model-Driven MCQ Generator for Personalized Learning," *IEEE Access*, vol. 12, pp. 102261-102273, 2024. doi:10.1109/ACCESS.2024.3420709
- [2] Q. Z. Lim, C. P. Lee, K. M. Lim, and A. K. Samangan, "UniRaG: Unification, Retrieval and Generation for Multimodal Question Answering With Pre-Trained Language Models," *IEEE Access*, vol. 12, pp. 71505–71512, May 2024. doi: 10.1109/ACCESS.2024.3403101
- [3] Babakhani, P., Lommatzsch, A., Brodt, T., Sacker, D., Sivrikaya, F., & Albayrak, S. "Opinerium: Subjective Question Generation Using Large Language Models," *IEEE Access*, vol. 12, pp. 66085-66092, May 2024. doi: 10.1109/ACCESS.2024.3398553
- [4] J. Xu, Y. Sun, J. Gan, M. Zhou, and D. Wu, "Leveraging Structured Information from a Passage to Generate Questions", *Tsinghua Science and Technology*, vol. 28, no. 3, pp. 464–474, Jun. 2023. doi: 10.26599/TST.2022.9010034
- [5] H. Yang, "A Study on an Intelligent Algorithm for Automatic Test Paper Generation and Scoring in University English Exams," *Journal of ICT Standardization*, vol. 11, no. 4, pp. 391-402, 2023. doi: 10.13052/JICTS2245-800X.1144
- [6] H. M. Balaha and M. M. Saafan, "Automatic Exam Correction Framework (AECF) for the MCQs, Essays, and Equations Matching," *IEEE Access*, vol. 9, pp. 32368-32372, Feb. 2021. doi: 10.1109/ACCESS.2021.3060940
- [7] A. Srivastava and A. Memon, "Toward Robust Evaluation: A Comprehensive Taxonomy of Datasets and Metrics for Open Domain Question Answering in the Era of Large Language

Models," IEEE Access, vol. 12, pp. 117483-117489, Sep. 2024. doi: 10.1109/ACCESS.2024.3446854

[8] H. G. Deokate, P. P. Jogdand, P. S. Satpute, and S. B. Shaikh, "Automatic Question Generation from Given Paragraph," International Journal for Scientific Research & Development (IJSRD), vol. 7, no. 3, pp. 1521-1523, Mar. 2019

[9] T. Steuer, A. Filighera, and T. Tregel, "Investigating Educational and Noneducational Answer Selection for Educational Question Generation," IEEE Access, vol. 10, pp. 63522-63536, Jun. 2022. doi: 10.1109/ACCESS.2022.3180838

[10] G. Kurdi, J. Leo, B. Parsia, U. Sattler, and S. Al-Emari, "A Systematic Review of Automatic Question Generation for Educational Purposes," International Journal of Artificial Intelligence in Education, vol. 30, pp. 121-147, 2019. doi: 10.1007/s40593-019-00186-y